```
pip install seaborn
```

```
Requirement already satisfied: seaborn in .\lib\site-packages (0.13.2)
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Requirement already satisfied: pandas>=1.2 in .\lib\site-packages (from seaborn) (2.2.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in .\lib\site-packages (from seaborn) (3.10.1)
Requirement already satisfied: contourpy>=1.0.1 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.2)
Requirement already satisfied: cycler>=0.10 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.57.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.8)
Requirement already satisfied: packaging>=20.0 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (25.0)
Requirement already satisfied: pillow>=8 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (11.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in .\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in .\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in .\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in .\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
pip install tensorflow
```

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Successfully installed absl-py-2.2.2 astunparse-1.6.3 certifi-2025.4.26 charset-normalizer-3.4.1 flatbuffers-25.2.10 gast-0.6.0 googl
Note: you may need to restart the kernel to use updated packages.
```

```
!pip install opencv-python
```

```
Requirement already satisfied: opencv-python in .\lib\site-packages (4.11.0.86)
Requirement already satisfied: numpy>=1.21.2 in .\lib\site-packages (from opencv-python) (2.1.2)
```

```
import cv2
print(cv2.__version__)
```

```
4.11.0
```

```
pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in .\lib\site-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in .\lib\site-packages (from scikit-learn) (2.1.2)
Requirement already satisfied: scipy>=1.6.0 in .\lib\site-packages (from scikit-learn) (1.15.2)
Requirement already satisfied: joblib>=1.2.0 in .\lib\site-packages (from scikit-learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in .\lib\site-packages (from scikit-learn) (3.6.0)
Note: you may need to restart the kernel to use updated packages.
```

```
import os
import pandas as pd
import numpy as np
import seaborn as sns
import cv2
from glob import glob
import tensorflow as tf
import matplotlib.pyplot as plt
from PIL import Image
from tensorflow.keras.preprocessing import image
from matplotlib.image import imread
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from keras.layers import BatchNormalization
from sklearn.metrics import classification_report,confusion_matrix
from tensorflow.keras.models import Sequential, Model
from keras.regularizers import l2
from tensorflow.keras.layers import Activation, Dropout, Dense, Flatten, Conv2D, BatchNormalization, MaxPooli
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.applications import InceptionV3
from tensorflow.keras.applications.inception_v3 import preprocess_input
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint, ReduceLROnPlateau

import warnings
warnings.filterwarnings("ignore")
```

```
import os
import shutil
import random

# Define dataset paths
folder_path = "E:/cse366_research/OCT17/OCT2017"

train_dir = os.path.join(folder_path, "E:/cse366_research/OCT17/OCT2017/train")
val_dir = os.path.join(folder_path, "E:/cse366_research/OCT17/OCT2017/val")
test_dir = os.path.join(folder_path, "E:/cse366_research/OCT17/OCT2017t/test")
```

```
pip install torch torchvision matplotlib tqdm
```

```
Requirement already satisfied: torch in .\lib\site-packages (2.6.0+cu124)
Requirement already satisfied: torchvision in .\lib\site-packages (0.21.0+cu124)
```

```
Requirement already satisfied: matplotlib in .\lib\site-packages (3.10.1)
Requirement already satisfied: tqdm in .\lib\site-packages (4.67.1)
Requirement already satisfied: filelock in .\lib\site-packages (from torch) (3.13.1)
Requirement already satisfied: typing-extensions>=4.10.0 in .\lib\site-packages (from torch) (4.12.2)
Requirement already satisfied: networkx in .\lib\site-packages (from torch) (3.3)
Requirement already satisfied: jinja2 in .\lib\site-packages (from torch) (3.1.4)
Requirement already satisfied: fsspec in .\lib\site-packages (from torch) (2024.6.1)
Requirement already satisfied: sympy==1.13.1 in .\lib\site-packages (from torch) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in .\lib\site-packages (from sympy==1.13.1->torch) (1.3.0)
Requirement already satisfied: numpy in .\lib\site-packages (from torchvision) (2.1.2)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in .\lib\site-packages (from torchvision) (11.0.0)
Requirement already satisfied: contourpy>=1.0.1 in .\lib\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in .\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in .\lib\site-packages (from matplotlib) (4.57.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\lib\site-packages (from matplotlib) (1.4.8)
Requirement already satisfied: packaging>=20.0 in .\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pyparsing>=2.3.1 in .\lib\site-packages (from matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in .\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: colorama in .\lib\site-packages (from tqdm) (0.4.6)
Requirement already satisfied: six>=1.5 in .\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: MarkupSafe>=2.0 in .\lib\site-packages (from jinja2->torch) (2.1.5)
Note: you may need to restart the kernel to use updated packages.
```

```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as T
from torchvision import models
import os
import pandas as pd
from torchvision.io import read_image
from torch.utils.data import Dataset
from torch.utils.data import DataLoader
import matplotlib.pyplot as plt
import time
from tqdm import tqdm
```

```
import torch
from torch.utils.data import Dataset
import os
from torchvision.io import read_image
import torchvision.transforms as T

class CustomImageDataset(Dataset):
    def __init__(self, ds_path, transforms=None, target_transform=None):
        self.ds_path = ds_path
        self.transforms = transforms
        self.target_transform = target_transform
        self.labels = os.listdir(ds_path)  # Get list of label folder names
        self.img_paths = []

        # Create a list of image paths and ensure all labels are mapped correctly
        for label in self.labels:
            base_path = os.path.join(ds_path, label)
            if os.path.isdir(base_path):  # Ensure it's a directory
                for img in os.listdir(base_path):
                    img_path = os.path.join(base_path, img)
                    self.img_paths.append(img_path)

        # Mapping labels to numeric values
        self.label_map = {

            'CNV': 0,

            'DME': 1,
```

```
            'DRUSEN': 2,

            'NORMAL': 3
        }

    def __len__(self):
        return len(self.img_paths)

    def __getitem__(self, idx):
        img_path = self.img_paths[idx]
        image = read_image(img_path)

        # Extract the label from the folder name in a cross-platform way
        label = os.path.basename(os.path.dirname(img_path))

        # Ensure label exists in label_map, otherwise raise an error
        if label not in self.label_map:
            raise KeyError(f"Label '{label}' not found in label_map. Check dataset structure.")

        image = image.float() / 255.0  # Normalize pixel values to [0,1]

        # Convert grayscale images to RGB if needed
        if image.shape[0] == 1:
            image = image.repeat(3, 1, 1)

        # Apply transformations
        if self.transforms:
            image = self.transforms(image)

        image = T.Resize((224, 224))(image)  # Ensure image size is consistent

        return image, self.label_map[label]
```

```
train_path = 'E:/cse366_research/OCT17/OCT2017/train'
train_data = CustomImageDataset(train_path)

val_path = 'E:/cse366_research/OCT17/OCT2017/val'
val_data = CustomImageDataset(val_path)

test_path = 'E:/cse366_research/OCT17/OCT2017/test'
test_data = CustomImageDataset(test_path)

train_len = len(train_data)
train_dataloader = DataLoader(train_data, batch_size=64, shuffle=True)

val_len = len(val_data)
val_dataloader = DataLoader(val_data, batch_size=64, shuffle=True)

test_len = len(test_data)
test_dataloader = DataLoader(test_data, batch_size=4)

data_loaders = {'train': train_dataloader, 'val': val_dataloader}
```

```
device = torch.device("cuda:0" if torch.cuda.is_available() else "cpu")
device
```

```
device(type='cuda', index=0)
```

```
def __getitem__(self, idx):
    img_path = self.img_paths[idx]
    image = read_image(img_path)

    # Extract the label correctly for both Windows and Linux
    label = os.path.basename(os.path.dirname(img_path))  # Correct way to get folder name

    image = image.squeeze()
    image = image.repeat(3, 1, 1)
    image = image / 255.0
    if self.transforms:
        image = self.transforms(image)
    if self.target_transform:
        label = self.target_transform(label)

    image = T.Resize((224, 224))(image)

    return image, self.label_map[label]



class ResNetLSTM(nn.Module):
    def __init__(self, num_classes=4, lstm_hidden_size=256, lstm_num_layers=1):
        super(ResNetLSTM, self).__init__()

        # Load pretrained ResNet18 model
        self.resnet = models.resnet18(pretrained=True)
        # Remove the final fully connected layer
        self.resnet = nn.Sequential(*list(self.resnet.children())[:-1])

        # LSTM to process the features from ResNet
        self.lstm = nn.LSTM(input_size=512,  # The ResNet output size is 512
                            hidden_size=lstm_hidden_size,
                            num_layers=lstm_num_layers,
                            batch_first=True)

        # Fully connected layer for classification
        self.fc = nn.Linear(lstm_hidden_size, num_classes)

    def forward(self, x):
        # Pass through ResNet (without the final classification layer)
        x = self.resnet(x)  # Shape: [batch_size, 512, 1, 1]

        # Flatten ResNet output for LSTM
        x = x.view(x.size(0), -1)  # Shape: [batch_size, 512]
        x = x.unsqueeze(1)  # Adding sequence length dimension: [batch_size, 1, 512]

        # Pass through LSTM
        lstm_out, (hn, cn) = self.lstm(x)  # lstm_out: [batch_size, seq_len, hidden_size]

        # We only need the output from the last LSTM step
        lstm_out = lstm_out[:, -1, :]  # Shape: [batch_size, hidden_size]

        # Final classification
        out = self.fc(lstm_out)  # Shape: [batch_size, num_classes]
        return out



model = ResNetLSTM(num_classes=4)
model.to(device)

# Criterion and Optimizer
criterion = nn.CrossEntropyLoss()
```

```
optimizer = optim.Adam(model.parameters(), lr=0.001)


class EarlyStopping:
    def __init__(self, patience=5, min_delta=0):
        self.patience = patience
        self.min_delta = min_delta
        self.counter = 0
        self.best_loss = None
        self.early_stop = False
        self.stopped_epoch = 0

    def on_epoch_end(self, epoch, val_loss):
        if self.best_loss is None:
            self.best_loss = val_loss
        elif val_loss < self.best_loss - self.min_delta:
            self.best_loss = val_loss
            self.counter = 0
        else:
            self.counter += 1
            if self.counter >= self.patience:
                self.early_stop = True
                self.stopped_epoch = epoch


def train_model(model, criterion, optimizer, data_loaders, device, num_epochs, patience=5):
    early_stop = EarlyStopping(patience=patience)

    history = {'train_loss': [], 'val_loss': [], 'train_acc': [], 'val_acc': []}

    for epoch in range(num_epochs):
        print(f'Epoch {epoch+1}/{num_epochs}')
        print('-' * 10)

        for phase in ['train', 'val']:
            if phase == 'train':
                model.train()
            else:
                model.eval()

            running_loss, running_corrects = 0.0, 0
            total = 0

            for inputs, labels in tqdm(data_loaders[phase]):
                inputs = inputs.to(device)
                labels = labels.to(device)

                with torch.set_grad_enabled(phase == 'train'):
                    outputs = model(inputs)
                    _, preds = torch.max(outputs, 1)
                    loss = criterion(outputs, labels)

                    if phase == 'train':
                        optimizer.zero_grad()
                        loss.backward()
                        optimizer.step()

                running_loss += loss.item() * inputs.size(0)
                running_corrects += torch.sum(preds == labels.data)
                total += inputs.size(0)

            epoch_loss = running_loss / total
            epoch_acc = running_corrects.double() / total
```

```
            history[f"{phase}_loss"].append(epoch_loss)
            history[f"{phase}_acc"].append(epoch_acc.item())

            print(f'{phase} Loss: {epoch_loss:.4f} Acc: {epoch_acc:.4f}')

            if phase == 'val':
                early_stop.on_epoch_end(epoch, epoch_loss)
                if early_stop.early_stop:
                    print("Early stopping triggered.")
                    return history

    return history
```

```
train_data = CustomImageDataset('E:/cse366_research/OCT17/OCT2017/train')
history=train_model(model, criterion, optimizer, data_loaders, device, num_epochs=5)
```

```
Epoch 1/5
----------
100%|██████████| 1305/1305 [05:34<00:00,  3.90it/s]
train Loss: 0.1409 Acc: 0.9523
100%|██████████| 1/1 [00:00<00:00,  3.69it/s]
val Loss: 0.0498 Acc: 1.0000
Epoch 2/5
----------
100%|██████████| 1305/1305 [05:24<00:00,  4.02it/s]
train Loss: 0.1178 Acc: 0.9599
100%|██████████| 1/1 [00:00<00:00,  4.84it/s]
val Loss: 0.0733 Acc: 1.0000
Epoch 3/5
----------
100%|██████████| 1305/1305 [05:30<00:00,  3.95it/s]
train Loss: 0.1006 Acc: 0.9656
100%|██████████| 1/1 [00:00<00:00, 12.97it/s]
val Loss: 0.0312 Acc: 1.0000
Epoch 4/5
----------
100%|██████████| 1305/1305 [05:47<00:00,  3.75it/s]
train Loss: 0.0857 Acc: 0.9704
100%|██████████| 1/1 [00:00<00:00, 12.21it/s]
val Loss: 0.1772 Acc: 0.9375
Epoch 5/5
----------
100%|██████████| 1305/1305 [05:45<00:00,  3.78it/s]
train Loss: 0.0732 Acc: 0.9745
100%|██████████| 1/1 [00:00<00:00, 13.69it/s]val Loss: 0.0383 Acc: 1.0000
```

```
def evaluate_accuracy(model, dataloader, device):
    model.eval()
    correct = 0
    total = 0

    with torch.no_grad():
        for images, labels in dataloader:
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            _, predicted = torch.max(outputs, 1)
            correct += (predicted == labels).sum().item()
            total += labels.size(0)

    accuracy = correct / total
    return accuracy
```

```
test_acc = evaluate_accuracy(model, test_dataloader, device)
print(f"Test Accuracy: {test_acc * 100:.2f}%")
```

```
Test Accuracy: 99.69%
```